

Name: [REDACTED]

Date: Monday, September 30, 2024

Total
18
/18

KNOWLEDGE

1. Divide the function $x^3 - 28x - 41$ by $x + 4$ using long division. Write the corresponding division statement.

$$\begin{array}{r}
 x^2 - 4x - 12 \\
 x+4 \overline{) x^3 + 0x^2 - 28x - 41} \\
 \underline{-x^2 + 4x^2} \\
 -4x^2 - 28x \\
 \underline{-4x^2 - 16x} \\
 -12x - 41 \\
 \underline{-12x - 48} \\
 7 = R
 \end{array}$$

division statement

$$x^3 - 28x - 41 = (x+4)(x^2 - 4x - 12) + 7$$

2. a) List the possible factors of $2x^3 - 9x^2 + 10x - 3$.

$$b: \pm 1, \pm 3 \quad b/a: \pm 1, \pm 3, \pm \frac{1}{2}, \pm \frac{3}{2}$$

$$a: \pm 1, \pm 2$$

possible factors: $(x-1), (x+1), (x-3), (x+3), (2x-1)(2x+1)$

- b) Fully factor the function $2x^3 - 9x^2 + 10x - 3$ using synthetic division. $(3x-2)(3x+2)$

$$f(1) = 0 \rightarrow (x-1)$$

$$\begin{array}{r|rrrr}
 1 & 2 & -9 & 10 & -3 \\
 & \downarrow & & & \\
 + & & 2 & 7 & 3 \\
 \hline
 x & 2 & -7 & 3 & 0
 \end{array}$$

$$2x^2 - 7x + 3$$

$$\begin{array}{r}
 1 \quad -3 \\
 2 \quad -1 \\
 \hline
 = (x-3)(2x-1)(x-1)
 \end{array}$$

$\therefore$ the factors of $2x^3 - 9x^2 + 10x - 3$ are $(x-3)(2x-1)(x-1)$ which are

$$(3, 0), \left(\frac{1}{2}, 0\right), (1, 0)$$

- ③ Given $8x^3 + 27$, solve for x . List all real and complex numbers (if any exist) in simplified form.

$$8x^3 + 27$$

$$= (2x+3)(4x^2 - 2x(3) + 9)$$

$$= (2x+3)(4x^2 - 6x + 9)$$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{6 \pm \sqrt{(-6)^2 - 4(4)(9)}}{2(4)}$$

$$= \frac{-6 \pm \sqrt{-108}}{8}$$

$$= \frac{6 \pm 6\sqrt{3}i}{8}$$

$$= \frac{3 \pm 3\sqrt{3}i}{4}$$

$\therefore$ the real + complex roots are

$$x_1 = (2x+3)$$

$$x_2 = \frac{3 + 3\sqrt{3}i}{4}$$

$$x_3 = \frac{3 - 3\sqrt{3}i}{4}$$

4. When a polynomial is divided by $(x - 2)$, its quotient is $x^2 - 3x - 7$ and its remainder is -24 . What is the polynomial?

$$\text{dividend} = \text{divisor} \cdot \text{quotient} + R$$

$$= (x-2)(x^2 - 3x - 7) - 24$$

$$= x^3 - 3x^2 - 7x - 2x^2 + 6x + 14 - 24$$

$$= x^3 - 5x^2 - x - 10$$

$\therefore x^3 - 5x^2 - x - 10$
is the polynomial

APPLICATION

1. Determine the value of k such that the remainder of $(2x^3 + 4x^2 - kx + 5) \div (x + 3)$ is the same as the remainder of $(6y^3 - 3y^2 + 2y + 7) \div (2y - 1)$.

$$\hookrightarrow x = \frac{1}{2}$$

$$\hookrightarrow x = -3$$

$$f\left(\frac{1}{2}\right) = 6\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right) + 7$$

$$= 8$$

$$8 = 2(-3)^3 + 4(-3)^2 - (-3)k + 5$$

$$8 = -13 + 3k$$

$$21 = 3k$$

$$k = 7$$

$\therefore$ the value of
 $k = 7$

2. The volume, in cubic centimeters, of a rectangular box can be modelled by $2x^3 + x^2 - 27x - 36$. Determine the possible dimensions of the box if the height, in centimetres, is given by $x + 3$.

$$\hookrightarrow x = -3$$

$$2x^3 + x^2 - 27x - 36$$

$$(x-4)(2x+3)(x+3)$$

$\therefore$ the possible dimensions of

the cube are $x = 4, -\frac{3}{2}, -3$

$(4, 0)(-\frac{3}{2}, 0)(-3, 0)$ but

only $x = 4$ is applicable because
in real life, dimensions cannot

< 0

$$\begin{array}{r|rrrrr} -3 & 2 & 1 & -27 & -36 & \\ & \downarrow & & & & \\ + & & -6 & 15 & 36 & \\ \hline \times & 2 & -5 & -12 & 0 & \end{array}$$

$$2x^2 - 5x - 12$$

$$\begin{array}{r} 1 \\ 2 \end{array} \quad \begin{array}{r} -4 \\ 3 \end{array}$$

$$(x-4)(2x+3)(x+3)$$

(4)

(3)

9